Q1: Find the period of $f(t)=(10\cos t)^2$ , using identity:-

Cosx*cosy= 0.5(cos(x-y) +cos(x+y))

Q2: A signal has a fundamental frequency of 1000 Hz.

What is the period in Nano seconds??

Q3: Express the following in their simplest form:

Sin $(2\pi ft)$ + sin $(2\pi ft-\pi)$